

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0611

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

*CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model,
(BL 6)*

Assessment Tool Weightage: 20%

QUESTIONS: 2

Total Marks:20

Annexure A

Case Study

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

Probabilistic Model Design:

- Use Bayes' Theorem to update disease probability given symptoms.
- Apply Naïve Bayes for fast classification assuming independence.
- Employ Bayesian Belief Networks (BBN) to capture dependencies (e.g., fever → infection → pneumonia).

Handling Uncertainty:

- Probabilities represent confidence, not absolutes.
- Missing data is managed by marginalization.
- Posterior probabilities guide diagnosis under uncertainty.

Conditional Dependencies:

- Naïve Bayes may overestimate when symptoms are correlated.
- BBN corrects bias by explicitly modeling dependencies.

Probability Learning:

- Parameters learned from patient records.
- Larger datasets improve accuracy; online learning adapts to new diseases.

Real-World Constraints:

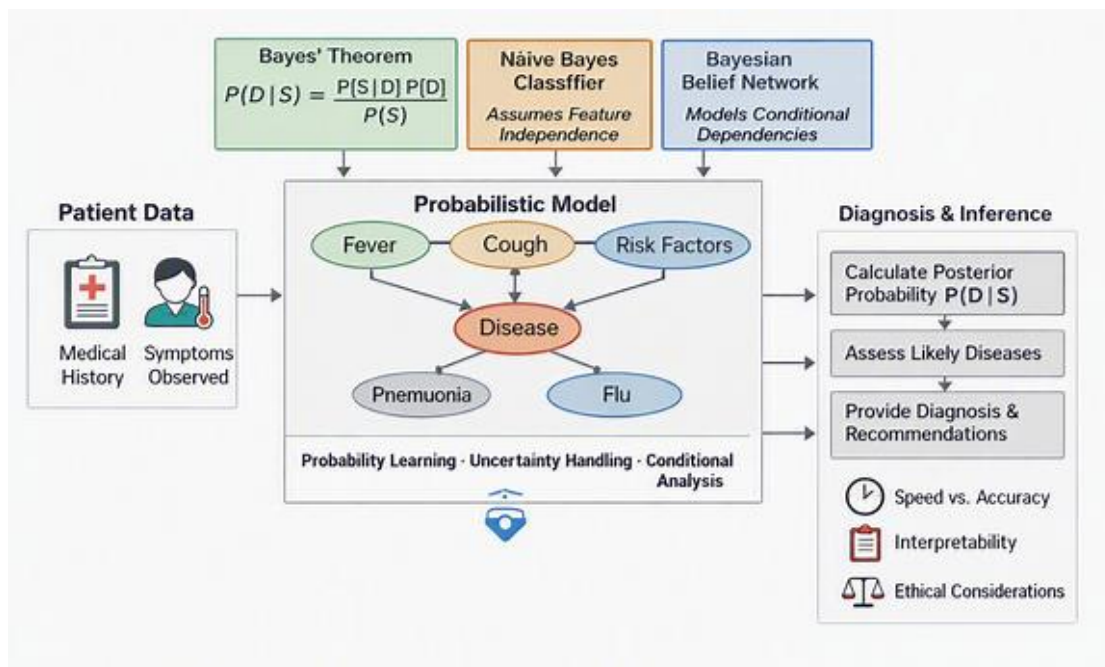
- Data quality issues, need for interpretability, scalability, and ethical use.
- Hybrid Naïve Bayes + BBN balances speed and accuracy.

Concept Overview: Bayesian models use probability to infer disease likelihood from symptoms and history. Bayes' Theorem updates prior belief $P(D)$ with evidence $P(S | D)$ to compute posterior $P(D | S)$.

Model Design:

- **Naïve Bayes Classifier:** assumes symptom independence; fast and effective for large datasets.
- **Bayesian Belief Network (BBN):** captures conditional dependencies (e.g., fever → infection → pneumonia).
- **Learning Process:** model parameters learned from patient data using Maximum Likelihood Estimation.

Uncertainty Handling: Probabilities quantify uncertainty; missing data handled via marginalization. BBN improves reliability by modeling symptom interrelations.



2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

- **Classification Model:** Naïve Bayes treats words as independent features (e.g., “free”, “win”). Bayes Optimal Classifier considers all hypotheses but is computationally heavy.
- **Feature Selection:** Maximum Likelihood (MLE): maximizes fit but risks overfitting. Minimum Description Length (MDL): balances accuracy with simplicity, reducing overfitting.
- **Performance Analysis:** Naïve Bayes is fast and effective for small datasets. Bayes Optimal is more accurate but impractical at scale.
- **Trade-offs in Complexity:** Complex models → better accuracy but slower. Simple models → scalable, interpretable, slightly less accurate.
- **Impact of Sample Size & Hypothesis Space:** Small samples favor Naïve Bayes. Large samples could benefit Bayes Optimal, but cost is high. MDL helps constrain hypothesis space to avoid overfitting.

Concept Overview: Spam detection applies Bayes’ Theorem to classify emails as spam or not spam based on word frequencies.

Model Design:

- **Naïve Bayes:** assumes independence among words; efficient for text classification.
- **Bayes Optimal Classifier:** theoretically best but computationally expensive.

Feature Selection:

- **Maximum Likelihood Estimation (MLE):** selects features maximizing probability fit.

- **Minimum Description Length (MDL):** prefers simpler models that compress data efficiently, reducing overfitting.

Performance & Trade-offs:

- Naïve Bayes → fast, interpretable, good for small datasets.
- Bayes Optimal → higher accuracy, but impractical for large hypothesis spaces.
- Larger sample size improves estimation; smaller hypothesis space enhances generalization.

Evaluation: Accuracy measured via precision, recall, and F1-score. Trade-off between complexity and interpretability determines real-world success.

